SGPMAIN6.0v.py: Specular Grassmann-Polarity Index Method

User's Manual and Conceptual Framework

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August 2026

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GLOSSARY OF TERMS

Symbol/Term	**Meaning**	**Page**
P^+	Positively charged amino acids (H, K, R)	13
P^-	Negatively charged amino acids (D, E)	13
N	Neutral polar amino acids (C, G, N, Q, S, T, Y)	13
NP	Nonpolar amino acids (A, F, I, L, M, P, V, W)	13
v	PIM vector of 16 dimensions	13
$v \wedge w$	Wedge product of v and w	15
$\mathrm{WedgeSim}(v, w)$	Corrected wedge similarity	15
$R_c(v)$	Specular reflection of v	16
$\Sigma(v)$	Clifford signature of v	17
$H(v)$	Shannon entropy of v	**??**
$\mathrm{JS}(v, w)$	Jensen-Shannon divergence	**??**
$G(v)$	Gini coefficient	**??**
$C(v)$	Structural complexity index	**??**
$I(v)$	Moran's I (spatial autocorrelation)	**??**
ΔG	Gibbs free energy (stability)	**??**
T_m	Melting temperature	**??**
$\rho(v, w)$	Spearman correlation	**??**
$H_d(v, w)$	Hellinger distance	**??**
λ_k	Karhunen-Loève eigenvalue	19
PIM	Polarity Index Method	13
GA	Grassmann Algebra	15
SGPMAIN6.0v.py	Specular Grassmann-Polarity Index Method v17	3
PIDP	Protein Intrinsic Disorder Prediction	7
GRAVY	Grand Average of Hydropathicity	**??**
pI	Isoelectric point	**??**
KL	Karhunen-Loève decomposition	19
IDP	Intrinsically Disordered Protein	7

Note: Page numbers correspond to the first mention of the term in the text. Terms in **bold** are key concepts of the SGPMAIN system.

Abstract

This document presents the complete user's manual and conceptual framework of the SGPMAIN6.0v.py (Specular Grassmann-Polarity Index Method) system, a computational platform for protein analysis based on three fundamental pillars: (1) the Polarity Index Method (PIM) as a representation of proteins as 16-dimensional vectors [1], (2) Grassmann Algebra as the mathematical engine for comparing and transforming polarity profiles [2, 3, 4], and (3) derived chemical properties that provide value to chemists, biochemists, and bioinformaticians. The system includes a Protein Intrinsic Disorder Prediction (PIDP) module that integrates metapredict and AIUPred for disorder analysis, a Karhunen-Loève decomposition module for dimensionality reduction, and a therapeutic peptide design module. All mathematical formulas from the LUJO manuscript are included with clear, accessible explanations of their biological and computational significance. Flow diagrams, property tables, and concrete examples using the Lujo virus (LUJO) envelope glycoprotein as a case study are included. This manual includes installation instructions, configuration guides, output file descriptions, and troubleshooting. The complete source code is available at `https://github.com/polancodf2024/lujv-sgp-analysis`.

Version Information: This manual corresponds to SGPMAIN6.0v.py, which processes 151,066,931 sequences across 26 groups in approximately 13 hours and 21 minutes.

User Manual

INSTALLATION AND SYSTEM REQUIREMENTS

Prerequisites

Before installing SGPMAIN6.0v.py, ensure your system meets the following requirements:

Table 2: System requirements for SGPMAIN6.0v.py

Component	**Minimum Requirement**
Operating System	Linux, macOS, or Windows (with WSL)
Python	Version 3.8 or higher
RAM	4 GB (16 GB recommended for large datasets)
Storage	10 GB free space
Processor	64-bit, 2 or more cores

Required Python Packages

The following Python packages are required. They are listed in `requirements.txt`:

```
numpy >= 1.21.0
pandas >= 1.3.0
scipy >= 1.7.0
scikit-learn >= 1.0.0
biopython >= 1.79
matplotlib >= 3.4.0
seaborn >= 0.11.0
psutil >= 5.8.0
\# Optional for PIDP
metapredict >= 1.0
aiupred >= 3.0
\# Optional for therapeutic profiler
chembl_webresource_client >= 2.0
```

Listing 1: Required packages

Installation Steps

1. Clone the repository:

```
git clone https://github.com/polancodf2024/lujv-sgp-analysis.git
cd lujv-sgp-analysis
```

2. Install dependencies:

```
pip install -r requirements.txt
```

3. (Optional) Install PIDP dependencies for intrinsic disorder prediction:

```
# For metapredict - machine learning-based disorder prediction
pip install metapredict

# For AIUPred - biophysical model-based disorder prediction
pip install aiupred

# Verify PIDP installation
python3 -c "import metapredict; import aiupred; print('PIDP packages OK')"
```

4. Verify the installation:

```
python3 SGPMAIN6.0v.py --help
```

PIDP Dependencies

For full PIDP functionality, the following additional packages are required:

Table 3: PIDP dependencies

Package	Version	Description
metapredict	¿= 1.0	Machine learning disorder prediction
aiupred	¿= 3.0	Biophysical disorder and redox sensitivity

Note: If PIDP packages are not installed, the program will skip PIDP analysis and continue with other analyses. This is not a fatal error.

Verification

To verify that the installation was successful, run the test script:

```
python3 -c "import numpy, pandas, scipy, sklearn; print('All packages imported successfully')"
```

INPUT FILE FORMATS

FASTA File Format

SGPMAIN6.0v.py accepts standard FASTA format files with the extension `.fasta` or `.dat0`:

```
>sp|P06935|VGP_WNV Envelope glycoprotein
MNRLV... (sequence continues)
>sp|Q05320|VGP_WNV Envelope glycoprotein
MNRLV... (sequence continues)
```

Listing 2: Example FASTA file format

FASTA Format Rules

- Each record starts with a header line beginning with the greater-than symbol
- The header contains the protein identifier and optional description
- Sequence lines follow the header (may span multiple lines)
- Only standard amino acid letters are allowed (A through Z, case-insensitive)

Required Input Files

Table 4: Input files required by SGPMAIN6.0v.py

File	**Format**	**Description**
*.unico.dat0	FASTA	Protein sequences for each group
chembl_uniprot.txt	TSV	ChEMBL-UniProt mapping (optional)
apd_natural.fasta	FASTA	Antiviral peptide database (optional)

Group File Naming Convention

Each group file should follow the naming pattern:

`{group_name}.unico.dat0`

For example:

- `lujo.unico.dat0` - Lujo virus envelope glycoprotein
- `lasv.unico.dat0` - Lassa virus glycoprotein
- `CPP.unico.dat0` - Cell-penetrating peptides

ChEMBL Mapping File (Optional)

The optional file `chembl_uniprot.txt` maps UniProt proteins to ChEMBL targets:

```
# chembl_37 target list, 01/05/2026
P21266  CHEMBL2242      Glutathione S-transferase Mu 3  SINGLE PROTEIN
P27658  CHEMBL3314      Collagen alpha-1(VIII) chain  SINGLE PROTEIN
```

If this file is not present, the TherapeuticProfiler will still work but will not include ChEMBL IDs in the output.

APD Peptide Database (Optional)

The optional file `apd_natural.fasta` contains antimicrobial peptides for training the activity prediction model:

```
>AP00001
GLWSKIKEVGKEAAKAAAKAAGKAALGAVSEAV
>AP00002
KISKAIAKLAKNKAEKAKAAAKAIKAAKEAKA
```

If this file is not present, the activity prediction model will not be trained.

Setting the Target Group

The target group is the primary protein or virus family to be analyzed. To change it, modify the `MAIN_GROUP` variable in the configuration section:

```python
# For Lujo virus (default in v17)
MAIN_GROUP = ['lujo']

# For West Nile virus
MAIN_GROUP = ['nile1', 'nile2']

# For Ebola
MAIN_GROUP = ['zaire', 'sudan', 'reston', 'bundibugyo', 'tai', 'bombali']

# For custom protein groups
MAIN_GROUP = ['your_protein_group_1', 'your_protein_group_2']
```

Listing 3: Configuring the target group

Key Configuration Parameters

Table 5: Key configuration parameters in SGPMAIN6.0v.py

Parameter	Default	Description
MAX_STORED_PROTEINS_PER_GROUP	10000	Maximum proteins stored per group
BATCH_SIZE	100000	Number of sequences per batch
PROGRESS_REPORT_INTERVAL	100000	Frequency of progress updates
USE_WEIGHTS	True	Enable biological weighting
USE_TRIPLETS	True	Enable tri-vector calculations
USE_QUADRUPLETS	True	Enable quadri-vector calculations
TOP_N_PROTEINS	20	Number of top proteins to return

PIDP Configuration Parameters

The PIDP module can be configured with the following parameters:

Table 6: PIDP configuration parameters

Parameter	Default	Description
USE_PIDP	True	Enable/disable PIDP analysis
PIDP_TARGETS_ONLY	True	Only analyze MAIN_GROUP targets
PIDP_USE_METAPREDICT	True	Use metapredict for disorder prediction
PIDP_USE_AIUPRED	True	Use AIUPred for disorder prediction
PIDP_THRESHOLDS	[0.3, 0.4, 0.5]	Disorder thresholds to report

Enabling and Disabling PIDP

To enable or disable PIDP analysis, modify these variables in the configuration:

```
# Enable/disable PIDP analysis
USE_PIDP = True  # Set to False to disable

# Choose which tools to use
PIDP_USE_METAPREDICT = True  # Use metapredict (if installed)
PIDP_USE_AIUPRED = True      # Use AIUPred (if installed)

# Disorder thresholds to report
PIDP_THRESHOLDS = [0.3, 0.4, 0.5]
```

Listing 4: PIDP configuration

Advanced Configuration Options

Table 7: Advanced configuration options

Parameter	**Default**	**Description**
USE_BIOLOGICAL_METRIC	True	Use biological metric signature
USE_GRASSMANN_PROJECTION	True	Enable Grassmann projection [5]
USE_FUBINI_STUDY	True	Enable Fubini-Study metric [6, 7]
USE_RICCI_CURVATURE	True	Enable Ricci curvature calculation
USE_KARHUNEN_LOEVE	True	Enable KL decomposition
USE_RADON_TRANSFORM	True	Enable Radon transform [8]
USE_FRACTAL_DIMENSION	True	Enable fractal dimension [9]
USE_WASSERSTEIN	True	Enable Wasserstein distance
USE_POLARITY_LAPLACIAN	True	Enable polarity Laplacian

Basic Usage

```
# Run the complete analysis
python3 SGPMAIN6.0v.py

# The results will be saved in:
# results_v17_YYYYMMDD_HHMMSS/
```

Listing 5: Basic execution

Example: Analyzing Lujo Virus (LUJO)

1. Ensure the following files are in the working directory:

```
lujo.unico.dat0
lasv_all.unico.dat0  junv_all.unico.dat0
macv_all.unico.dat0  lcmv_all.unico.dat0
CPP.unico.dat0  NON_CPP.unico.dat0
UNFOLDED.unico.dat0  PARTIALLY_FOLDED.unico.dat0
REVIEWED_HUMAN.unico.dat0  UNREVIEWED_HUMAN.unico.dat0
REVIEWED_ALL.unico.dat0  UNREVIEWED_ALL.unico.dat0
VIRUS_REVIEWED.unico.dat0  VIRUS_UNREVIEWED.unico.dat0
SIGNALS.unico.dat0  MEMBRANE.unico.dat0  DISEASE.unico.dat0
```

2. Set `MAIN_GROUP` in the configuration:

```
MAIN_GROUP = ['lujo']
```

3. Run the analysis:

```
python3 SGPMAIN6.0v.py
```

4. Monitor the progress:

```
[PROGRESS] Progress [UNREVIEWED_ALL]:
149,232,295 sequences | Valid: 149,232,294 (100.0%) |
Rate: 3,128 seq/s | ETA: calculating...
[MEMORY] Memory: 500 MB | Stored: 10000 (sample)
```

Example Output

After a successful run, you will see:

```
[SUCCESS] Comparison saved:
results_v17_20260812_095649/comparison_lujo_vs_all.csv

[SUCCESS] Similarity matrix saved:
results_v17_20260812_095649/similarity_matrix_groups.csv

[SUCCESS] Top 20 individual proteins saved:
results_v17_20260812_095649/top_20_individual_proteins.csv

[SUCCESS] Therapeutic profile saved:
results_v17_20260812_095649/therapeutic_profile.json

[SUCCESS] Chemical analysis saved for LUJO
[SUCCESS] Chemical profile saved: chemical_profile_lujo.csv (54 properties)

[SUCCESS] PIDP analysis saved: pidp_analysis_lujo.csv
[SUCCESS] PIDP summary saved: pidp_summary_all_targets.csv
[SUCCESS] PIDP peptide analysis saved: pidp_peptide_analysis.csv

[SUCCESS] KL decomposition saved: kl_decomposition.json
[SUCCESS] Target group statistics saved: target_groups_stats_v17.csv
```

Example PIDP Output

When PIDP analysis is enabled and the tools are installed, you will see output like:

```
[PROTEIN] Performing PIDP analysis on target proteins...
[PROTEIN] PIDP TOOLS STATUS:
     [SUCCESS] metapredict: Available (v3.0.2)
     [SUCCESS] aiupred: Available (v3.x)
[PROTEIN] LUJO: metapredict: 8.15%, AIUPred: 9.69%
[PROTEIN] Synthetic peptide: metapredict: 100.0%, AIUPred: 45.45%
[SUCCESS] PIDP analysis saved: pidp_analysis_lujo.csv
[SUCCESS] PIDP summary saved: pidp_summary_all_targets.csv
[SUCCESS] PIDP peptide analysis saved: pidp_peptide_analysis.csv
```

Performance Statistics from LUJO Run

The LUJO run processed the following:

Table 8: Performance statistics from LUJO run (August 12, 2026)

Metric	Value
Total sequences read	151,066,931
Total valid PIMs	151,066,930
Total rejected	1
Validity rate	100.00%
Total bytes processed	72.17 GB
Processing time	13 hours 21 minutes
Average speed	3,141 seq/s
Peak memory usage	500 MB
Number of groups	26

Performance Optimization

For large datasets, consider these optimizations:

Table 9: Performance optimization strategies

Strategy	How to implement
Reduce sample size	Set MAX_STORED_PROTEINS_PER_GROUP = 1000
Disable plots	Set GENERATE_PLOTS = False
Disable bootstrap	Set USE_BOOTSTRAP = False
Reduce components	Set USE_TRIPLETS = False
Disable PIDP	Set USE_PIDP = False
Reduce batch size	Set BATCH_SIZE = 50000

Summary of Output Files

File	Description
comparison_{target}_vs_all.csv	Wedge similarity between target and all groups
similarity_matrix_groups.csv	Cross-group similarity matrix (26×26)
top_20_individual_proteins.csv	Top 20 individual proteins most similar to target
chemical_profile_{group}.csv	Complete chemical profile (54 properties in one file)
therapeutic_profile.json	Antiviral peptide design with activity prediction
kl_decomposition.json	Karhunen-Loève decomposition (eigenvalues, explained variance)
target_groups_stats_v17.csv	Statistics of target groups (cohesion, threshold, entropy, etc.)
pidp_analysis_{group}.csv	Intrinsic disorder prediction per target
pidp_summary_all_targets.csv	PIDP summary for all targets
pidp_peptide_analysis.csv	PIDP analysis for synthetic peptide

PIDP Output Files

pidp_analysis_{group}.csv

This file contains the intrinsic disorder prediction results for each target protein using metapredict and AIUPred.

Table 11: Columns in pidp_analysis_{group}.csv

Column	Description
Tool	Prediction tool (metapredict or aiupred)
Metric	Disorder threshold (0.3, 0.4, 0.5) or statistics (mean, max, min, std)
Value	Percentage of disordered residues or statistical value

Example for LUJO:

```
Tool,Metric,Value
metapredict,disorder_0.3,12.33
metapredict,disorder_0.4,9.91
metapredict,disorder_0.5,8.15
metapredict,mean_score,0.1655
metapredict,max_score,1.0
metapredict,min_score,0.02
metapredict,std_score,0.2248
aiupred,disorder_0.3,20.26
aiupred,disorder_0.4,12.78
aiupred,disorder_0.5,9.69
aiupred,mean_score,0.1919
aiupred,max_score,0.8846
aiupred,min_score,0.0317
aiupred,std_score,0.1894
```

pidp_summary_all_targets.csv

This file compares PIDP results across all target proteins and the synthetic peptide.

Table 12: Columns in pidp_summary_all_targets.csv

Column	Description
Protein/Peptide	Name of the protein or peptide
Length	Sequence length
Is Peptide	True/False
metapredict_disorder_0.3	Percent disordered at threshold 0.3 (metapredict)
metapredict_disorder_0.4	Percent disordered at threshold 0.4 (metapredict)
metapredict_disorder_0.5	Percent disordered at threshold 0.5 (metapredict)
metapredict_mean	Mean disorder score (metapredict)
aiupred_disorder_0.3	Percent disordered at threshold 0.3 (AIUPred)
aiupred_disorder_0.4	Percent disordered at threshold 0.4 (AIUPred)
aiupred_disorder_0.5	Percent disordered at threshold 0.5 (AIUPred)
aiupred_mean	Mean disorder score (AIUPred)
aiupred_redox	Redox sensitivity (AIUPred)

Example:

```
Protein/Peptide,Length,Is Peptide,metapredict_disorder_0.3,metapredict_disorder_0.4,...
lujo,454,False,12.33,9.91,8.15,0.1655,20.26,12.78,9.69,0.1919,0.029
synthetic_peptide,11,True,100.0,100.0,100.0,0.7249,45.45,45.45,45.45,0.4114,0.0
```

pidp_peptide_analysis.csv

This file contains the PIDP analysis specifically for the synthetic peptide designed by the TherapeuticProfiler.

Example:

```
Tool,Metric,Value
metapredict,disorder_0.3,100.0
metapredict,disorder_0.4,100.0
metapredict,disorder_0.5,100.0
metapredict,mean_score,0.7249
metapredict,max_score,0.7692
metapredict,min_score,0.6894
metapredict,std_score,0.0284
aiupred,disorder_0.3,45.45
aiupred,disorder_0.4,45.45
aiupred,disorder_0.5,45.45
aiupred,mean_score,0.4114
aiupred,max_score,0.9355
aiupred,min_score,0.0024
aiupred,std_score,0.4485
```

KL Decomposition Output

kl_decomposition.json

This file contains the Karhunen-Loève decomposition of the covariance matrix:


```
{
  "eigenvalues": [
    0.012774646436720669,
    0.009003358767749265,
    0.004103228058453734,
    0.0034442317518499216,
    0.0025955642745731987,
    0.0018834140679670395,
    0.001480897143947259,
    0.0010622415498677312
  ],
  "explained_variance": [
    0.3249281781847145,
    0.22900398664177232,
    0.10436722646799369,
    0.08760539510178646,
    0.06601920258819298,
```

```
    0.047905380779296916,
    0.03766720382010455,
    0.027018533413065024
  ],
  "n_components": 8
}
```


Interpreting KL Decomposition

Table 13: Karhunen-Loève decomposition interpretation

Component	**Interpretation**
Eigenvalues	Show the variance captured by each component
Explained Variance	Percentage of total variance explained
First 8 components	Capture 92.45% of total variance

Interpreting PIDP Results

Comparison of PIDP Tools

Table 14: Comparison of PIDP tools

Feature	**metapredict**	**AIUPred**
Method	Machine Learning (Random Forest)	Biophysical model
Input	Sequence + PSSM + Secondary Structure	Sequence only
Redox sensitivity	No	Yes
Speed	Fast	Moderate
Accuracy	AUC 0.89	AUC 0.87

Practical Interpretation of PIDP Results

- **Low disorder (0 to 10%)**: Protein is predominantly ordered. Suitable for crystallization and structure determination.
- **Moderate disorder (10 to 30%)**: Protein has flexible regions that may be important for function.
- **High disorder (> 30%)**: Protein is largely unstructured. Consider using intrinsically disordered protein (IDP) analysis methods.

For the LUJO glycoprotein, the low disorder (8.15% at threshold 0.5 for metapredict; 9.69% at threshold 0.5 for AIUPred) confirms that it is a predominantly ordered protein, consistent with its role as a structural glycoprotein involved in receptor binding and membrane fusion.

Note: PIDP results should be interpreted in conjunction with other properties. A protein can have low disorder but still contain functionally important flexible regions.

Interpreting the Comparison File

The comparison file contains the following key columns:

Table 15: Columns in the comparison file

Column	**Description**
Compared Group	Name of the comparison group
Wedge Similarity	Higher values indicate greater similarity (0 to 1 scale)
Wedge Orientation	Positive or negative sign
Probability of Belonging	Statistical probability of group membership
Hydrophobic Angle	Rotation angle in the hydrophobic plane (degrees)
Charge Angle	Rotation angle in the charge plane (degrees)
Specular Reflection	True/False for charge inversion detection
Clifford Distance	Distance in Clifford algebra
Commutator Norm	Norm of the commutator
Anticommutator Sim	Anticommutator similarity
Grassmann Distance	Distance in Grassmann space
Fubini-Study	Fubini-Study metric
Ricci Curvature	Ricci curvature
Wasserstein Distance	Wasserstein-1 distance
Fractal Dim Diff	Fractal dimension difference
Radon Similarity	Radon transform similarity
Laplacian Distance	Laplacian distance
Entropy Diff	Entropy difference
Jensen-Shannon	Jensen-Shannon divergence
Gini Diff	Gini coefficient difference
Spearman	Spearman correlation
Morans I Diff	Moran's I difference

Interpreting Chemical Properties

Table 16: Key chemical properties and their interpretation

Property	Range	Interpretation
Net Charge	-3 to +3	Positive = basic protein, Negative = acidic protein
pI	3 to 10	pH at which net charge is zero
GRAVY	-4.5 to +4.5	Negative = hydrophilic, Positive = hydrophobic
Solubility	1 to 20 mg/mL	Higher = more soluble
Delta G	-15 to -2 kcal/mol	More negative = more stable
Tm	40 to 80 °C	Higher = more thermally stable
PIDP disorder	0 to 100%	Higher = more intrinsically disordered
Glycosylation sites	N-linked, O-linked	Number of predicted sites

How to Present Results in Papers

Similarity Tables

When presenting similarity results, include:

1. The target group
2. Wedge similarity values (with 6 decimal places)
3. Orientation signs
4. Hydrophobic and charge angles
5. Number of samples in each group

Example table format from the LUJO analysis:

Table 17: Top groups most similar to LUJO glycoprotein

Group	**Wedge Similarity**	**Hydro. Angle**	**Charge Angle**
CPP	0.1223	26.45°	29.44°
NON_CPP	0.0738	28.51°	27.51°
NILE1	0.0672	15.30°	30.26°
UNFOLDED	0.0522	12.92°	11.98°

Reporting Chemical Properties

When reporting chemical properties, include:

- Net charge at physiological pH (7.4)
- Isoelectric point (pI)
- GRAVY score
- Predicted stability (Delta G)
- Number of glycosylation sites
- Number of phosphorylation sites
- Lipid binding preferences

Example from the LUJO analysis:

```
The Lujo virus glycoprotein has a net charge of +0.052 at pH 7.4,
a pI of 5.97, and a GRAVY score of 0.253 (slightly hydrophobic).
The predicted folding stability is -6.76 kcal/mol with a melting
temperature of 41.04 degrees Celsius. The protein has 12 N-linked
glycosylation sites, 64 O-linked sites, 32 phosphorylation sites,
and 7 acetylation sites. The best lipid binding is cholesterol
(score 0.554), followed by ganglioside (0.343) and
phosphatidylserine (0.417).
```

Reporting PIDP Results

When reporting intrinsic disorder prediction results, include:

```
Intrinsic disorder analysis using metapredict and AIUPred revealed
that the LUJO glycoprotein has low disorder propensity with
8.15% of residues predicted as disordered at threshold 0.5 by
metapredict and 9.69% by AIUPred. The mean disorder score was
0.1655 for metapredict and 0.1919 for AIUPred. The AIUPred redox
sensitivity was 0.029, indicating minimal redox-dependent structural
changes.
```

Reporting KL Decomposition

When reporting KL decomposition results, include:

```
Karhunen-Loève decomposition revealed that the first component
explains 32.49% of the total variance, while the first eight
components cumulatively explain 92.45% of the variance, suggesting
that the polarity profile follows a structured, low-dimensional
organization.
```

Reporting Hot Spots

From the LUJO chemical profile:

```
Three drug-binding hot spots were identified in the LUJO glycoprotein:
1. Hydrophobic patch (fusion loop-like): Score 0.339,
   Residues: D512, E515, H516, I518, W519, A520
2. Mixed polar/hydrophobic (binding loop-like): Score 0.268,
   Residues: F134, F136, Y139, D142, K144
3. Charge-rich region (CR2-like): Score 0.104,
   Residues: H54, E64, D66, K68, H71
```

Reporting Therapeutic Peptide Design

When reporting therapeutic peptide design, include:

```
A competitor peptide (NDLKNAAAAAA) was designed with predicted
antiviral activity score of 0.853 (confidence: 0.933), estimated
affinity of 0.012 nM, net charge of 0.0, molecular weight of
1209.3 Da, hydrophobicity of 0.018, isoelectric point of 6.0,
and solubility of 14.96 mg/mL in PBS buffer.
```

Citing SGPMAIN6.0v.py

When citing SGPMAIN6.0v.py in your publications, use the following format:

```
Polanco, C., Uversky, V.N., et al. (2026). SGPMAIN6.0v.py: Specular
Grassmann-Polarity Index Method for Protein Functional
Characterization. Bulletin of Mathematical Biology, 88(12), 1-45.
```

Steps to Add a New Protein Group

1. Prepare a FASTA file: `your_group.unico.dat0`
2. Add the group name to `files_to_load` in `main()`:

```
files_to_load = {
    ...
    'your_group': 'your_group.unico.dat0',
    ...
}
```

3. Add the group to `MAIN_GROUP` if it should be a target group:

```
MAIN_GROUP = ['your_group']
```

4. (Optional) Add the group to `GROUP_NAME_MAP` for display:

```
GROUP_NAME_MAP = {
    'your_group': 'YOUR_GROUP_DISPLAY_NAME',
}
```

Example: Adding a New Virus Strain

```
# 1. Add to files_to_load
files_to_load = {
    ...
    'new_virus': 'new_virus.unico.dat0',
    ...
}

# 2. Add to MAIN_GROUP
MAIN_GROUP = ['new_virus', 'lujo']

# 3. Add to GROUP_NAME_MAP
GROUP_NAME_MAP = {
    'new_virus': 'NEW_VIRUS_STRAIN',
}
```

Common Issues and Solutions

Issue	Solution
File not found: *.unico.dat0	Verify the file exists in the working directory
MemoryError	Reduce MAX_STORED_PROTEINS_PER_GROUP
No valid PIM vectors	Check sequence length (greater than 2 amino acids)
Slow execution	Reduce BATCH_SIZE or use fewer groups
LaTeX compilation errors	Use the provided user_manual.tex file
Python import errors	Install missing packages with pip
Permission denied	Use sudo or change file permissions
NameError: GrassmannPIM not defined	Ensure class ordering is correct in SGP-MAIN6.0v.py
PIDP tools not available	Install metapredict or aiupred
KL decomposition fails	Check covariance matrix is positive definite

PIDP-Specific Issues

Table 19: PIDP-specific troubleshooting

Issue	Solution
metapredict not installed	Install: pip install metapredict
aiupred not installed	Install: pip install aiupred
PIDP analysis skipped	Check USE_PIDP = True in configuration
No sequence for PIDP	Ensure sequences are stored in sample_data
PIDP results missing	Check that target proteins are in MAIN_GROUP
AIUPred redox profile fails	Check aiupred version (¿= 3.0 required)

Performance Optimization

- For large datasets, set MAX_STORED_PROTEINS_PER_GROUP to 1000
- Disable GENERATE_PLOTS for faster execution
- Use USE_BOOTSTRAP = False for faster comparisons
- Reduce N_BOOTSTRAP from 100 to 50 or 20
- Disable PIDP if not needed: USE_PIDP = False
- Reduce BATCH_SIZE to 50000 for memory-constrained systems

Error Messages and Solutions

Table 20: Common error messages and solutions

Error Message	**Solution**
File not found	Check file path and name
MemoryError	Reduce sample size or use fewer groups
No valid PIM vectors	Check sequence quality and length
ImportError: No module named...	Install missing package with pip
Segmentation fault	Reduce BATCH_SIZE or use smaller dataset
NameError: name 'GrassmannPIM' is not defined	Move GrassmannPIM class before AdvancedGroupAnalyzer
PIDP: metapredict not available	Install metapredict or set PIDP_USE_METAPREDICT to False
PIDP: aiupred not available	Install aiupred or set PIDP_USE_AIUPRED to False
Covariance matrix singular	Check for duplicate sequences in group

QUICK REFERENCE CARD

Key Functions

Table 21: Key functions in SGPMAIN6.0v.py

Function	Description
compute_pim_profile(seq)	Convert sequence to PIM vector [1]
wedge_product(v, w)	Calculate wedge similarity
clifford_signature(v)	Calculate Clifford signature [2, 3]
analyze_protein(group)	Full chemical analysis
get_top_individual_proteins(target, n)	Get top N proteins
PIDPProfiler.analyze_sequence(seq, name)	Analyze intrinsic disorder
PIDPProfiler.analyze_target_proteins(dir)	Analyze all target proteins
compute_kl_decomposition(group)	Karhunen-Loève decomposition
design_therapeutic_peptide(target)	Design competitor peptide

Command Summary

```
# Run full analysis
python3 SGPMAIN6.0v.py

# Check help
python3 SGPMAIN6.0v.py --help

# View results
ls results_v17_*/

# Open summary
cat results_v17_*/chemical_profile_lujo.csv

# Open comparison
cat results_v17_*/comparison_lujo_vs_all.csv

# View PIDP results
cat results_v17_*/pidp_summary_all_targets.csv

# View KL decomposition
cat results_v17_*/kl_decomposition.json

# View therapeutic profile
```

```
cat results_v17_*/therapeutic_profile.json
```

Listing 6: Quick commands

File Locations

Table 22: Default file locations

File Type	Location
Input files	Current working directory
Output files	results_v17_YYYYMMDD_HHMMSS/
Log files	results_v17_YYYYMMDD_HHMMSS/log.txt
Configuration	SGPMAIN6.0v.py (edit directly)

Important Notes

1. Always backup your input files before running
2. Results are timestamped to prevent overwriting
3. Large datasets may require several hours to process
4. The program supports Ctrl+C for safe interruption
5. Logs are saved for debugging and reproducibility
6. PIDP requires metapredict and/or aiupred to be installed
7. The complete source code is available at: `https://github.com/polancodf2024/lujv-sgp-analysis`
8. Version 17 includes corrected wedge similarity (more stringent than previous versions)

Part I

Conceptual Framework and System Architecture

CHAPTER 1

INTRODUCTION TO THE SGPMAIN6.0V.PY SYSTEM

1.1 Overview and Motivation

The SGPMAIN6.0v.py (Specular Grassmann-Polarity Index Method) system is a computational platform for protein analysis that moves beyond traditional sequence alignment methods to capture the physicochemical essence of proteins. The central insight is that protein function is fundamentally determined by the spatial and temporal organization of polarity transitions along the polypeptide chain, not by the primary sequence of amino acids per se. This insight is crucial because two proteins with completely different sequences can have similar functions if they share similar polarity patterns, and conversely, proteins with similar sequences can have different functions if their polarity patterns diverge.

The system is built on three fundamental pillars that work together to provide a comprehensive analysis framework:

1. **The Polarity Index Method (PIM)**: This is the foundational representation layer. It transforms the complex, high-dimensional space of amino acid sequences into a compact 16-dimensional vector space where each dimension corresponds to a specific polarity transition. This transformation is mathematically rigorous yet biologically interpretable, as each transition directly corresponds to a specific physicochemical interaction.

2. **Grassmann Algebra**: This is the computational engine that operates on the PIM vectors. Grassmann algebra provides a natural mathematical framework for comparing, transforming, and analyzing vector subspaces. The key insight is that the wedge product of two PIM vectors gives a measure of their geometric relationship that is both mathematically sound and biologically meaningful.

3. **Derived Chemical Properties and PIDP**: These are the outputs that connect the abstract mathematics to practical biochemistry. By applying carefully designed transformations to the PIM vector, we can extract properties such as net charge, isoelectric point, hydrophobicity, stability, glycosylation sites, phosphorylation sites, drug-binding hot spots, and intrinsic disorder prediction. These properties are directly interpretable and actionable for experimental scientists.

1.2 What's New in Version 17

SGPMAIN6.0v.py includes the following new features and improvements:

1. **Corrected Wedge Similarity**: A more stringent similarity measure that produces values closer to 0 for orthogonal profiles, improving discriminative power.

2. **Karhunen-Loève Decomposition**: Dimensionality reduction that identifies the most informative components of the polarity profile.

3. **Enhanced PIDP Integration**: Full integration of metapredict and AIUPred with redox sensitivity profiles.

4. **Chemical Profiler: Unified Output**: All 54 chemical properties in a single CSV file per group.

5. **Therapeutic Peptide Design**: Machine learning-based prediction of antiviral activity for designed competitor peptides.

6. **18 Mathematical Metrics**: Shannon entropy, Jensen-Shannon divergence, Gini coefficient, Moran's I, Hellinger distance, Spearman correlation, structural complexity index, functional modularity index, and more.

7. **Improved Performance**: Processing of 151,066,931 sequences in 13 hours and 21 minutes at 3,141 seq/s.

1.3 System Architecture: Three-Level Design

The SGPMAIN6.0v.py system operates through a carefully designed three-level architecture that transforms raw sequence data into actionable biochemical insights. Each level builds upon the previous one, creating a coherent pipeline from sequence to function.

Level 1: Polarity Index Method (PIM) - The Foundation

At the most fundamental level, SGPMAIN6.0v.py implements the Polarity Index Method (PIM). This level takes as input the amino acid sequence of a protein and produces as output a 16-dimensional vector. The transformation is accomplished through the following steps:

First, each amino acid in the sequence is classified into one of four polarity categories based on its physicochemical properties. These categories are: positively charged (P^+: H, K, R), negatively charged (P^-: D, E), neutral polar (N: C, G, N, Q, S, T, Y), and non-polar hydrophobic (NP: A, F, I, L, M, P, V, W). This classification captures the essential electrostatic and hydrogen-bonding characteristics of each residue.

Second, the sequence is scanned to count the frequency of each possible transition between polarity categories. There are 16 possible transitions (four categories to four categories), and these counts form the components of the PIM vector.

Third, the vector is normalized so that all components sum to one, producing a probability distribution that is independent of sequence length. This normalization ensures that proteins of different lengths can be compared directly.

The result of Level 1 is a compact, length-independent representation of the protein's polarity organization that captures the essential physicochemical information needed for functional analysis. This representation is the foundation upon which all subsequent analyses are built.

Level 2: Grassmann Algebra - The Mathematical Engine

At the intermediate level, SGPMAIN6.0v.py applies Grassmann Algebra to operate on the PIM vectors produced by Level 1. Grassmann algebra provides a natural mathematical framework for working with vector subspaces and their geometric relationships [2, 3].

The key operations at this level include:

1. The wedge product, which computes the area of the parallelogram spanned by two PIM vectors. This provides a measure of their geometric relationship that captures both similarity and orthogonality.

2. The corrected wedge similarity, which normalizes the wedge product to produce a score between 0 and 1. A score of 1 indicates identical polarity profiles, while a score of 0 indicates orthogonal profiles.

3. Specular reflection, which tests for charge-inverted functional equivalence by inverting the charge components of the PIM vector.

4. The Clifford signature, which combines multiple geometric and information-theoretic measures into a comprehensive fingerprint of the protein's polarity profile.

5. Karhunen-Loève decomposition, which identifies the most informative components of the polarity profile for dimensionality reduction.

The result of Level 2 is a set of geometric measures that describe the relationships between proteins in terms of their polarity organization. These measures are mathematically rigorous and biologically interpretable.

Level 3: Derived Chemical Properties and PIDP - The Biochemical Outputs

At the highest level, SGPMAIN6.0v.py transforms the abstract vector space of PIM vectors into concrete biochemical properties that are directly interpretable by experimental scientists. This level comprises three complementary modules:

Chemical Properties Module: This module derives over 20 physicochemical properties from the PIM vector, including:

- Net charge at physiological pH, which determines electrostatic behavior
- Isoelectric point (pI), the pH at which the protein has zero net charge
- GRAVY (Grand Average of Hydropathicity), a measure of overall hydrophobicity
- Solubility in PBS buffer, which is critical for experimental work
- Conformational stability (ΔG), a measure of folding free energy
- Melting temperature (T_m), indicating thermal stability

- Glycosylation sites (N-linked and O-linked), important for immune evasion
- Phosphorylation sites, indicating regulation by host kinases
- Drug-binding hot spots, potential targets for therapeutic intervention
- Reactivity profile (cysteine, lysine, and tyrosine accessibility)
- Metal binding sites (calcium and zinc)
- Membrane permeability and lipid binding preferences
- Buffer stability recommendations

Protein Intrinsic Disorder Prediction (PIDP) Module: This module provides complementary structural information using two independent methods:

- metapredict: A machine learning-based predictor that combines sequence features, evolutionary information, and secondary structure predictions. It produces per-residue disorder scores and percentages of disordered residues at multiple thresholds.
- AIUPred: A biophysical model-based predictor that estimates disorder propensity based on amino acid composition and sequence context. It also provides redox sensitivity profiles, indicating how disorder changes under reducing versus oxidizing conditions.

Therapeutic Peptide Design Module: This module designs competitor peptides targeting the most similar human protein group:

- Identifies critical polarity transitions between target and host proteins
- Designs peptide sequences that mimic these transitions
- Predicts antiviral activity using a RandomForestRegressor trained on 3,306 antimicrobial peptides
- Estimates binding affinity
- Compares with known inhibitors
- Generates synthesis and formulation recommendations

The result of Level 3 is a comprehensive biochemical profile that connects the abstract mathematics of the previous levels to the practical needs of experimental scientists. This includes actionable information for protein purification, formulation, structural studies, and therapeutic design.

Summary of the Three-Level Architecture

Table 1.1: Summary of the three-level architecture of SGPMAIN6.0v.py

Level	**Input**	**Output**
Level 1: PIM	Amino acid sequence	16-dimensional PIM vector
Level 2: Grassmann Algebra	PIM vectors	Geometric measures (wedge similarity, Clifford signature, KL decomposition)
Level 3: Derived Properties	PIM vectors	Chemical properties + PIDP disorder prediction + Therapeutic peptides

This three-level architecture ensures that SGPMAIN6.0v.py is both mathematically rigorous and practically useful. The abstract mathematics at Levels 1 and 2 provide the foundation for robust, reproducible analysis, while the concrete outputs at Level 3 connect directly to the needs of experimental scientists.

Key Insight: The three levels form a coherent pipeline from sequence to function.
Level 1 transforms sequence into representation.
Level 2 analyzes relationships between representations.
Level 3 translates analysis into actionable biochemical insights.

1.4 The Protein Intrinsic Disorder Prediction (PIDP) Module

Motivation and Biological Significance

Protein intrinsic disorder is a fundamental property that has been increasingly recognized as crucial for understanding protein function. Intrinsically disordered proteins and regions (IDPs/IDRs) lack stable three-dimensional structure under physiological conditions yet perform essential functions such as molecular recognition, signaling, and regulation. The importance of disorder in viral proteins is particularly noteworthy, as many viral surface proteins contain disordered regions that facilitate immune evasion, receptor binding, and membrane fusion.

The PIDP module addresses the critical need for accurate disorder prediction in the SGPMAIN6.0v.py workflow. By integrating two complementary disorder prediction methods—metapredict and AIUPred—the module provides a robust assessment of protein disorder that can be integrated with the polarity and chemical profiling analysis. This integration is essential because disorder and polarity are fundamentally connected: disordered regions typically have distinct polarity signatures that distinguish them from ordered regions.

metapredict: Machine Learning-Based Disorder Prediction

metapredict [10] employs a machine learning approach that combines multiple sequence features to predict disorder propensity. The algorithm processes an input sequence through the following stages:

First, the amino acid sequence is encoded using one-hot encoding, which represents each residue as a binary vector of length 20. Second, evolutionary information is incorporated through a position-specific scoring matrix (PSSM) derived from sequence homology searches. Third, secondary structure predictions from a neural network are included as additional features. Finally, a Random Forest classifier with 500 decision trees integrates all these features to produce a per-residue disorder score.

The output of metapredict is a per-residue disorder score in the range from 0 to 1, where scores above 0.5 typically indicate disorder. The model was trained on a curated dataset of proteins with experimentally validated disorder, achieving an area under the ROC curve (AUC) of 0.89.

The integration of evolutionary information (PSSM) is particularly important, as it allows metapredict to capture patterns of conservation that are characteristic of ordered versus disordered regions. This makes it especially effective for analyzing viral proteins, which often have complex evolutionary histories.

AIUPred: Biophysical Model-Based Disorder Prediction

AIUPred [11] takes a fundamentally different approach, employing a biophysical model based on amino acid composition and sequence context. The algorithm is rooted in the observation that disorder propensity is largely determined by the balance of charged, polar, and hydrophobic residues.

The AIUPred algorithm models disorder as a combination of intrinsic residue propensities and context-dependent effects. Each residue contributes a position-specific bias term, and neighboring residues contribute context-specific weights. This approach captures both the local amino acid preferences for disorder and the influence of the surrounding sequence environment.

A key feature of AIUPred is its ability to predict redox sensitivity profiles. The algorithm can generate two separate disorder profiles: one for reducing conditions (redox-plus) and one for oxidizing conditions (redox-minus). The redox sensitivity is then defined as the difference between these two profiles, averaged over the sequence. This redox sensitivity is particularly relevant for viral proteins, as many are exposed to oxidative stress during infection and may undergo redox-dependent conformational changes.

Integration with the SGPMAIN6.0v.py Workflow

The PIDP module is seamlessly integrated into the SGPMAIN6.0v.py workflow through the PIDPProfiler class. The integration occurs after the PIM and chemical profiling stages, allowing for a complete characterization that includes disorder information. The key design principles are:

1. Compatibility: The PIDP module operates on the same sequence data that is used for PIM calculation, ensuring consistency between analyses.

2. Complementarity: Disorder analysis provides information that is orthogonal to polarity and chemical profiling, filling gaps in structural understanding.

3. Actionability: PIDP results are presented in a format that directly supports experimental design and interpretation.

Part II

Mathematical Description of Metrics and Algorithms

CHAPTER 2

THE POLARITY INDEX METHOD (PIM): MATHEMATICAL FOUNDATIONS

2.1 Introduction to the Mathematical Framework

The Polarity Index Method (PIM) is built on a rigorous mathematical foundation that transforms the complex, discrete space of amino acid sequences into a continuous vector space. This transformation is essential because it enables the application of powerful mathematical tools—linear algebra, differential geometry, and information theory—to the analysis of protein sequences. The key insight is that the physicochemical properties of proteins can be captured by the frequencies of transitions between polarity categories, which are independent of the specific amino acid identities.

The mathematical framework has several important properties that make it suitable for protein analysis:

1. Well-posedness: The transformation is uniquely defined for any sequence of sufficient length.
2. Continuity: Small changes in the sequence result in small changes in the PIM vector.
3. Completeness: The transformation captures all information about polarity transitions that is relevant to protein function.
4. Invariance: The transformation is invariant under permutations of the sequence that preserve the transition frequencies.

Definition 2.1.1 (Amino Acid Polarity)**.** *Let $\mathcal{A} = \{A_1, \ldots, A_{20}\}$ be the set of the 20 standard amino acids. We define the polarity function $\phi : \mathcal{A} \to \{P^+, P^-, N, NP\}$ as [1]:*

$$\phi(A) = \begin{cases} P^+ & \text{if } A \in \{H, K, R\} \\ P^- & \text{if } A \in \{D, E\} \\ N & \text{if } A \in \{C, G, N, Q, S, T, Y\} \\ NP & \text{if } A \in \{A, F, I, L, M, P, V, W\} \end{cases} \tag{2.1}$$

Definition 2.1.2 (PIM Vector)**.** *For a protein sequence $S = s_1 s_2 \ldots s_L$ of length L, we define the PIM vector $v \in \mathbb{R}^{16}$ as:*

$$v_i = \frac{1}{L-1} \sum_{k=1}^{L-1} \mathbf{1}\{\phi(s_k) = a, \phi(s_{k+1}) = b\} \tag{2.2}$$

where i is the index corresponding to the transition $a \to b$ in the order of the 16 possible transitions.

Example 2.1.1. *Let's compute the PIM vector for a simple protein sequence:* `AKDR`*. First, we classify each amino acid:*

- *A (Alanine) $\to$ NP (Nonpolar)*
- *K (Lysine) $\to$ P^+ (Positively charged)*
- *D (Aspartic acid) $\to$ P^- (Negatively charged)*
- *R (Arginine) $\to$ P^+ (Positively charged)*

The polarity string is: `NP`, P^+, P^-, P^+
Now we count the transitions:

- *NP $\to$ P^+: 1 transition*
- *$P^+ \to P^-$: 1 transition*
- *$P^- \to P^+$: 1 transition*

Normalizing by the total number of transitions ($L-1=3$), we get the PIM vector $v \in \mathbb{R}^{16}$:
The indices for the transitions are:

- *NP $\to$ P^+: index 12 (0-based counting: NP=3, P+=0, so $3 \times 4 + 0 = 12$)*
- *$P^+ \to P^-$: index 1 ($0 \times 4 + 1 = 1$)*
- *$P^- \to P^+$: index 4 ($1 \times 4 + 0 = 4$)*

Thus:

$$v_{12} = 1/3, \quad v_1 = 1/3, \quad v_4 = 1/3 \tag{2.3}$$

and all other components are 0.

Solución: *The PIM vector is:*

$$v = \begin{bmatrix} 0 & 1/3 & 0 & 0 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/3 & 0 & 0 & 0 \end{bmatrix}^T \tag{2.4}$$

□

CHAPTER 3

GRASSMANN ALGEBRA: FUNDAMENTAL OPERATIONS

3.1 Introduction to Grassmann Algebra

Grassmann algebra, also known as exterior algebra, provides a natural mathematical framework for working with vector subspaces and their geometric relationships [2, 3]. The fundamental operation in Grassmann algebra is the wedge product, which generalizes the cross product to arbitrary dimensions. This is essential for protein analysis because it allows us to measure the geometric relationship between two PIM vectors in a way that captures both their similarity and their orthogonality.

The key insight of Grassmann algebra is that while vectors represent points or directions, the wedge product of two vectors represents a plane (or more generally, a subspace). The magnitude of the wedge product gives the area of the parallelogram spanned by the vectors, and its direction gives the orientation of that parallelogram. This geometric interpretation is particularly powerful for protein analysis because it allows us to think of PIM vectors as directions in a 16-dimensional space, and the wedge product as a measure of how aligned those directions are.

3.2 Wedge Product and Corrected Similarity

Mathematical Definition and Interpretation

The wedge product of two PIM vectors captures the geometric relationship between them. The full wedge product in 16 dimensions would have 120 components, but we use a truncated version that focuses on the six biologically most relevant pairs of transitions. This truncation is justified because the six selected pairs capture the essential physicochemical information needed for functional comparison.

Definition 3.2.1 (Truncated Wedge Product)**.** *Let $v, w \in \mathbb{R}^{16}$ be two PIM vectors. We define the truncated wedge product over the set of biologically relevant pairs $\mathcal{K}$:*

$$\mathcal{K} = \{(0,5),(1,4),(2,6),(3,7),(10,11),(14,15)\} \tag{3.1}$$

The truncated wedge product is:

$$\omega_{\mathcal{K}}(v,w)_k = v_{i_k}w_{j_k} - v_{j_k}w_{i_k} \quad \textit{for } (i_k, j_k) \in \mathcal{K} \tag{3.2}$$

Definition 3.2.2 (Corrected Wedge Similarity)**.** *The corrected wedge similarity between two PIM vectors $v, w \in \mathbb{R}^{16}$ is defined as:*

$$\text{WedgeSim}(v, w) = 1 - \frac{\|\omega_{\mathcal{K}}(v, w)\|_2}{\|v\|_2 \|w\|_2} \tag{3.3}$$

This similarity satisfies $0 \leq \text{WedgeSim}(v, w) \leq 1$*, where* $\text{WedgeSim}(v, w) = 1$ *indicates identical vectors and* $\text{WedgeSim}(v, w) = 0$ *indicates orthogonal vectors.*

Remark 3.2.1. *The corrected wedge similarity is a more stringent measure than the uncorrected wedge similarity used in previous versions. A value of 0.12 indicates near-orthogonality of the polarity profiles, representing a substantial functional difference between the compared proteins. This improved discriminative power is the key advance of version 17.*

3.3 Specular Reflection

The Concept of Charge Inversion

Specular reflection is a novel concept introduced in SGPMAIN6.0v.py that captures the idea of charge-inverted functional equivalence. The insight is that two proteins might have similar functions even if their charge patterns are inverted—a phenomenon that occurs in some cases of molecular mimicry and immune evasion. The mathematical operation of specular reflection inverts the charge components of the PIM vector, allowing us to test whether two proteins are related by charge inversion.

The mathematical properties of specular reflection ensure that it is a well-defined geometric transformation:

1. The reflection is an involution (applying it twice returns the original vector).
2. The reflection preserves the Euclidean norm (it is an isometry).
3. The reflection is self-adjoint (it preserves the inner product structure).

Definition 3.3.1 (Reflection Normal Vector)**.** *The normal vector* $n \in \mathbb{R}^{16}$ *for specular reflection is defined as [12]:*

$$n_i = \begin{cases} +1 & \textit{if } i \in \{0, 1, 4, 5\} \textit{ (charge components)} \\ 0 & \textit{otherwise} \end{cases} \tag{3.4}$$

normalized as $n = n/\|n\|_2$*.*

Definition 3.3.2 (Specular Reflection)**.** *The specular reflection of a PIM vector* $v \in \mathbb{R}^{16}$ *is defined as:*

$$R_c(v) = v - 2(v \cdot n)n \tag{3.5}$$

This transformation inverts the sign of the charge components, simulating a charge inversion (P^+ *to* P^- *and* P^- *to* P^+*).*

3.4 Clifford Signature

Comprehensive Characterization

The Clifford signature provides a comprehensive characterization of a protein's polarity profile by combining multiple geometric, information-theoretic, and statistical measures into a single vector [2, 3]. Each component of the signature captures a different aspect of the polarity profile, and together they form a fingerprint that can be used for classification, comparison, and functional prediction.

The components of the Clifford signature are chosen to be:

1. Informative: Each component provides unique information about the polarity profile.
2. Interpretable: Each component has a clear biological meaning.
3. Stable: Each component is robust to small changes in the sequence.
4. Discriminative: Each component helps to distinguish between different functional classes of proteins.

Definition 3.4.1 (Clifford Signature)**.** *The Clifford signature of a PIM vector $v \in \mathbb{R}^{16}$ is the 12-component vector:*

$$\begin{aligned}\Sigma(v) = \big(&\|v\|,\ \mathrm{AutoRef}(v),\ \mathrm{HydroProj}(v),\ \mathrm{ChargeProj}(v),\ \mathrm{AutoRot}(v),\\ &H(v),\ D_f(v),\ G(v),\ C(v),\ M(v),\ I(v),\ \Lambda(v)\big)\end{aligned} \tag{3.6}$$

where:

$$\mathrm{AutoRef}(v) = \mathrm{WedgeSim}(v, R_c(v)) \tag{3.7}$$

$$\mathrm{HydroProj}(v) = |v_{10} + v_{15}| \tag{3.8}$$

$$\mathrm{ChargeProj}(v) = |v_0 + v_5| \tag{3.9}$$

$$\mathrm{AutoRot}(v) = \mathrm{WedgeSim}(v, \mathit{roll}(v, 4)) \tag{3.10}$$

CHAPTER 4

KARHUNEN-LOÈVE DECOMPOSITION

4.1 Dimensionality Reduction

The Karhunen-Loève decomposition (also known as Principal Component Analysis in the continuous domain) is applied to the covariance matrix of the polarity profile. This decomposition identifies the most informative components of the profile and provides a low-dimensional representation that preserves the essential information.

Definition 4.1.1 (Karhunen-Loève Decomposition)**.** *Let $\{v_1, \ldots, v_n\}$ be a set of PIM vectors. The covariance matrix is:*

$$C = \frac{1}{n-1} \sum_{i=1}^{n} (v_i - \bar{v})(v_i - \bar{v})^T \tag{4.1}$$

where $\bar{v}$ is the mean vector.

The Karhunen-Loève decomposition solves the eigenvalue problem:

$$C\phi_k = \lambda_k \phi_k, \quad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{16} \geq 0 \tag{4.2}$$

The explained variance for component k is:

$$\mathrm{EV}_k = \frac{\lambda_k}{\sum_{j=1}^{16} \lambda_j} \tag{4.3}$$

Example 4.1.1. *For the LUJO glycoprotein, the Karhunen-Loève decomposition revealed:*

Table 4.1: Karhunen-Loève decomposition for LUJO

Component	**Eigenvalue**	**Explained Variance (%)**
1	0.012775	32.49
2	0.009003	22.90
3	0.004103	10.44
4	0.003444	8.76
5	0.002596	6.60
6	0.001883	4.79
7	0.001481	3.77
8	0.001062	2.70
Cumulative		92.45

The rapid decay of eigenvalues confirms that the polarity profile follows a structured, low-dimensional organization.

APPENDIX A

REFERENCE DATABASE DESCRIPTION

A.1 UniProt Groups (v17)

The reference dataset consists of 26 groups totaling 151,066,931 sequences:

Group Code	Description	N
lujo	Lujo virus (target)	1
lasv	Lassa virus	4
junv	Junín virus	3
macv	Machupo virus	3
lcmv	Lymphocytic choriomeningitis virus	3
sudan	Ebola Sudan	1
zaire	Ebola Zaire	1
reston	Ebola Reston	1
bombali	Ebola Bombali	1
bundibugyo	Ebola Bundibugyo	1
tai	Ebola Taï Forest	1
nile1	West Nile virus membrane protein M	1
nile2	West Nile virus envelope glycoprotein E	1
CPP	Cell-Penetrating Peptides	100
NON_CPP	Non-Cell-Penetrating Peptides	71
UNFOLDED	Intrinsically Disordered Proteins	106
PARTIALLY_FOLDED	Partially Folded Proteins	153
REVIEWED_HUMAN	Reviewed Human Proteome	20,416
UNREVIEWED_HUMAN	Unreviewed Human Proteome	127,090
VIRUS_REVIEWED	Reviewed Virus Proteome	17,517
VIRUS_UNREVIEWED	Unreviewed Virus Proteome	1,087,163
REVIEWED_ALL	Reviewed All Proteome	575,503
UNREVIEWED_ALL	Unreviewed All Proteome	149,234,636
SIGNALS	Signaling Proteins	47
MEMBRANE	Membrane Proteins	165
DISEASE	Disease-Associated Proteins	3,942

A.2 LUJO Chemical Profile Summary

The chemical profile for LUJO includes the following key properties:

Table A.2: Selected chemical properties for LUJO

Property	**Value**
Net Charge	0.052
Charge Density	0.085
Charge Balance	0.808
pI	5.974
GRAVY	0.253
Hydrophobicity Score	0.196
N-Glycosylation Sites	12
O-Glycosylation Sites	64
Phosphorylation Sites	32
Acetylation Sites	7
Solubility (mg/mL)	12.81
ΔG (kcal/mol)	-6.760
Tm (°C)	41.04
Stability Score	0.552
Calcium Binding Score	0.303
Zinc Binding Score	0.450
Membrane Affinity	0.831
Best Lipid Binding	Cholesterol
Optimal pH	6.974
Salt Tolerance (mM)	241.50
Buffer Recommendation	PBS pH 7.0 + 241 mM NaCl + 9.2% glycerol
Storage Temperature	-20°C

A.3 APD Database

The Antimicrobial Peptide Database (APD) contains 3,306 antiviral peptides used for training the activity prediction model.

A.4 PIDP Tools

- metapredict: `https://github.com/idptools/metapredict`
- AIUPred: `https://github.com/idptools/aiupred`

A.5 Performance Statistics

Table A.3: Performance statistics for full run

Metric	**Value**
Total sequences read	151,066,931
Total valid PIMs	151,066,930
Total rejected	1
Validity rate	100.00%
Total bytes processed	72.17 GB
Processing time	13 hours 21 minutes
Average speed	3,141 seq/s
Peak memory usage	500 MB
Number of groups	26

A.6 Version History

Table A.4: Version history of SGPMAIN

Version	**Key Features**
v17	Corrected wedge similarity, KL decomposition, PIDP integration, therapeutic peptide design
v16	Grassmann projection, Fubini-Study metric, Ricci curvature
v15	Fractal dimension, Wasserstein distance, polarity Laplacian
v14	Radon transform, entropy-based metrics
v13	Specular reflection, Householder transformations
v12	Biological weighting, tri-vector calculations

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